

Name: Elton Fuentes Date: _____ Class: _____

MyDesign Portfolio

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 (HIGH Priority)

The organizer must securely lock the area to prevent students from taking supplies without permission and reduce theft or loss of classroom materials.

Previous solutions, such as open rolling carts, hanging pocket organizers, stacking drawers, and tabletop caddies, all failed because they had no locking mechanism, allowing supplies to go missing or be taken easily.

I shared this requirement with classmates. They agreed and said, “Yes, a lock is essential because supplies disappear during recess and lunch.”

Design Requirement #2 (HIGH Priority)

The organizer must have clearly labeled, separate compartments so students can find needed materials in under 30 seconds.

All prior designs, like the IKEA cart, shoe organizer, stacking drawers, and desk caddies suffered from supplies being mixed up or hard to locate quickly, causing lost instructional time.

My classmates reviewed this and confirmed, “We waste too much time searching — clear labels and sections are very important.” They suggested using both words and pictures for younger students. I added picture labels to the requirement.

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Design Requirement #3 (HIGH Priority)

The organizer must include a simple check-out/check-in system, such as a dry-erase chart, so teachers can track who took which supplies.

None of the previous solutions had any accountability or tracking method, which is why materials continued to go missing even when organized.

I told this to two of my classmates. They said the check-out chart is a great idea because it teaches responsibility. Students liked that it was simple. No major changes.

Design Requirement #4 (HIGH Priority)

The organizer must be made of durable materials that can withstand daily use by 20–30 elementary, middle school, or high school students without breaking or cracking within one school year.

Many existing plastic caddies and fabric organizers showed signs of damage or wear quickly when used by multiple students.

Some of my classmates validated this and said, “It has to be tough — kids are rough with supplies.” They recommended using thick plastic bins.

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Design Requirement #5 (HIGH Priority)

The organizer must be cheap to build or purchase so it is affordable for most classroom teachers or schools, and could be built easily and in bulk.

Prior solutions like commercial rolling carts and specialty caddies were often too expensive for teachers who already spend hundreds of dollars on supplies each year.

I asked some of my classmates. They agreed that anything too expensive would be unrealistic for most classrooms, and it would be hard to be multiple.

Design Requirement #6 (LOW Priority)

The organizer should be portable or have wheels so the teacher can easily move it between different areas of the classroom.

The IKEA rolling cart was popular mainly because of its mobility, even though it lacked security.

Some students liked the idea of mobility but said security and cost were more important. They noted it would be nice but not essential.

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Design Requirement #7 (LOW Priority)

The organizer should have a clean, colorful appearance with attractive labels to encourage students to use and return supplies properly.

Many prior designs that looked plain or messy were less likely to be used consistently by students.

Students I spoke with said they would be more careful with supplies if the organizer looked nice. My classmates agreed it would help, but ranked it as a lower priority than security and organization.

Rubric

Element C: Presentation and Justification of Solution Design Requirements						
C1. Determination of design requirements.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
_____ design requirements are related to prior research.	80%-100% of	60%-79% of	40%-59% of	20%-39% of	1%-19% of	No
C2. Requirements list.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
Design requirements _____.	are listed, prioritized, clear, and detailed	are listed, prioritized, generally clear, and detailed	are listed, prioritized, generally clear, and somewhat detailed	are listed and prioritized but some or all may be incomplete and/or lack specificity	are listed, formatted, and prioritization is attempted, but these may be partial or	are not present or are too vague

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					overly general	
C3. Quantity of high-priority design requirements. _____ of the design requirements listed are identified as high priority.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	Five	Four	Three	Two	One	None
C4. Objective & measurable. Design requirements _____.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	are consistently objective and measurable	are nearly always objective and measurable	are generally objective and measurable	may be only sometimes objective and/or measurable	are insufficiently measurable	are not measurable
C5. Leading to solution. Design requirements _____.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	would be highly likely to lead to a tangible and viable problem solution	would be likely to lead to a tangible and viable problem solution	have a potential to lead to a tangible and viable problem solution	have some potential to lead to a tangible and viable problem solution	have little potential to lead to a tangible and viable problem solution	cannot lead to a tangible and viable problem solution
C6. Primary stakeholder validation. There is evidence that requirements represent the needs of, and have been validated by, _____ primary stakeholder groups.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	80%-100% of	60%-79% of	40%-59% of	20%-39% of	1%-19% of	no primary stakeholder groups.